

FTC Autonomous Teacher Lesson Plan

Part 9 — Drive Forward Three Feet and Stop

Middle and high school FTC students | 60 to 75 minutes | Teams of up to eight

Students move from driver-controlled code to their first encoder-based autonomous mission. They calculate a theoretical 36-inch target, trace the program's independent stop conditions, perform the POWERING ON call-and-clear sequence, and use repeated measurements to calibrate the actual robot.

Lesson outcome

Students can explain and safely test the complete path from inches, to encoder counts, to `RUN_TO_POSITION`, to zero motor power. They distinguish a calculated distance from a guaranteed physical distance and support calibration with evidence.

Learning objectives

- Distinguish an Autonomous OpMode from a TeleOp.
- Calculate wheel circumference, counts per inch, and a 36-inch encoder target.
- Trace setup, `waitForStart`, target assignment, `RUN_TO_POSITION`, monitoring, and stop.
- Explain the purpose of `opModeIsActive`, `isBusy`, timeout, zero power, and BRAKE.
- Perform a call-and-clear sequence for a robot that will move across the floor.
- Measure repeated trials and calculate a calibration update.

Required materials

- [] 2026-2027 two-motor goBILDA StarterBot base with both drive encoder cables connected
- [] Control Hub, Driver Station, charged robot battery, and `StarterBotDriveForwardThreeFeet.java`
- [] At least six feet of clear floor: 36-inch travel lane plus start and stopping buffers
- [] Painter's tape, 12-foot tape measure, straightedge or square, and floor-safe marker labels
- [] Stable safety blocks for direction and encoder checks before floor motion
- [] Printed student workbook, focus guide, answer key, and POWERING ON poster
- [] A consistent robot reference point for every distance measurement

Teacher preparation

- [] Inspect drivetrain, wheel retention, wiring, battery, and encoder connectors.
- [] Confirm hardware names are exactly left_drive and right_drive.
- [] Confirm 19.2:1 motors, 96 mm drive wheels, and 1:1 motor-to-wheel reduction; edit constants if the build differs.
- [] Build the Java file and confirm the Autonomous OpMode appears on the Driver Station.
- [] Tape a straight start line, 36-inch target line, side boundaries, and an additional stopping buffer.
- [] Choose and mark the robot reference point used for both start and finish measurements.
- [] Perform a wheels-up direction test before class; do not pre-calibrate away the students' measurement task.

Starting calculation

ITEM	VALUE	MEANING
Motor encoder	537.7 counts/output revolution	19.2:1 Yellow Jacket specification
Drive wheel	96 mm = 3.7795 in	StarterBot wheel diameter
Drive reduction	1.0	Confirm motor-to-wheel ratio on the actual build
Wheel circumference	$3.7795 \times \pi = 11.874$ in	Distance per wheel revolution
Counts per inch	$537.7 / 11.874 = 45.2848$	Theoretical starting value
Target counts	$36 \times 45.2848 = 1,630.25$	Rounded to 1,630 whole counts

Critical distinction. The code commands approximately 1,630 motor counts. It does not measure 36 inches on the floor. Slip, wheel compression, alignment, and surface can change actual travel.

Student roles

ROLE	RESPONSIBILITY
Programmer	Finds the matching Java statement and records any approved constant change.
Safety captain	Leads POWERING ON, controls the boundary, and authorizes INIT/START.
Driver Station operator	Selects the OpMode and presses INIT, START, or STOP only when directed.

ROLE	RESPONSIBILITY
Start-line marshal	Places the disabled robot squarely at the taped line, then exits the lane.
Distance measurer	Measures from the same robot reference point after STOP and DISABLED.
Encoder recorder	Records target and final left/right counts from telemetry.
Trial recorder	Records commanded distance, actual distance, surface, and observations.
Mechanical observer	Watches straightness and consistency from outside the boundary.

Nonnegotiable safety boundary

No person may stand in the travel lane, stopping buffer, or beside the robot while it is enabled. A robot that has stopped moving may still be active. Press STOP, visually confirm DISABLED, and receive the safety captain's permission before measuring, repositioning, or touching it.

POWERING ON for floor motion

1. Place the robot squarely at the start while the Driver Station is DISABLED.
2. Remove hands, tools, cables, and measuring equipment from the lane.
3. Check the full lane, side boundaries, and stopping buffer.
4. Call loudly: POWERING ON!
5. Pause while everyone moves behind the boundary.
6. Visually confirm the complete area is clear.
7. Safety captain authorizes INIT, then START; operator remains ready to press STOP.

Video pause guide

PAUSE	TEACHER QUESTION	EXPECTED IDEA
0:09	What replaces the joystick in autonomous?	A prewritten sequence plus sensor feedback and stopping conditions.
0:18	What is our complete first mission?	Drive forward 36 inches, then command zero power.
0:29	Why must one drive motor be reversed?	Mirrored motors need opposite software directions to agree on physical forward.
0:39	What does an encoder measure—and	Motor rotation; it does not directly

PAUSE	TEACHER QUESTION	EXPECTED IDEA
	what does it not measure?	know floor position or wheel slip.
0:49	How does wheel circumference connect counts to inches?	One wheel revolution travels one circumference, so counts per revolution can be divided by inches per revolution.
0:58	Why is the target about 1,630 rather than exactly 1,630?	The calculation produces 1,630.25 and the motor target must be a whole count.
1:09	What happens between POWERING ON and START?	Pause, step behind the boundary, recheck the lane, and receive the safety captain's clear.
1:29	What three commands begin the move?	Set target positions, select RUN_TO_POSITION, and apply low positive power.
1:39	Why do we need STOP and a timeout in addition to encoder targets?	They provide independent exits if the run is canceled or the mechanism cannot reach its target.
1:49	Does BRAKE make an enabled robot safe to approach?	No. It reduces coasting; STOP and confirmed DISABLED are still required.
2:00	Why measure several trials?	Real distance varies; repeated evidence supports calibration and reveals mechanical inconsistency.

Lesson flow

STEP	ACTIVITY	TIME	TEACHER ACTION	STUDENT ACTION
1	Hook and predict	5 min	Ask how the robot could know it traveled three feet without a joystick.	Students propose time, rotation, or distance methods and compare limitations.
2	Watch with pauses	12 min	Use the pause guide and keep the robot disabled.	Students annotate the focus guide and explain each stop condition.
3	Calculate the target	10 min	Model circumference and counts per inch, then withhold the final product.	Students calculate and round the 36-inch target.

STEP	ACTIVITY	TIME	TEACHER ACTION	STUDENT ACTION
4	Trace the code	8 min	Assign setup, target, motion, monitor, and stop sections to groups.	Groups put code events in execution order and report their purpose.
5	Wheels-up validation	8 min	Run the safety sequence and verify encoder counts and physical forward at low power.	Students observe outside the boundary and record evidence.
6	Measured floor trials	15 min	Authorize one trial at a time; require STOP and DISABLED before measurement.	Students rotate roles and complete at least three trials.
7	Calibrate and retest	10 min	Approve one counts-per-inch update only after checking mechanical consistency.	Students calculate the update, predict the effect, and run a confirmation trial.
8	Debrief	5 min	Ask which evidence shows a code issue versus a mechanical issue.	Students complete the exit response.

Hands-on walkthrough

Build audit while disabled

Students identify motor model, wheel diameter, gear path, exact hardware names, and both encoder cables. They initial each fact rather than assuming the example matches.

Teacher question: Which constant would change if a gear reduction were added between motor and wheel?

Wheels-up direction and encoder test

Secure the robot on blocks. Run POWERING ON. Start the OpMode, stay outside the boundary, and verify both encoder values move toward positive targets while the wheels agree on physical forward. Press STOP and confirm DISABLED.

Teacher question: What mismatch would suggest a direction setting or encoder wiring problem?

First floor trial

The marshal squares the disabled robot on the start line. The measurer removes the tape from the lane. The safety captain clears the area and authorizes START. After motion ends, the operator presses STOP and confirms DISABLED before anyone enters.

Teacher question: What exact reference point will we measure so every trial is comparable?

Evidence before editing

Record target counts, final counts, actual distance, straightness, surface, and any wheel slip. If trials vary widely, inspect mechanics before changing a constant.

Teacher question: Would one short trial prove the counts-per-inch value is wrong? Why not?

Calibration and confirmation

Average consistent trial distances. Use new CPI = old CPI times commanded distance divided by actual distance. Change only the CPI value, rebuild, repeat the full safety sequence, and run a confirmation trial.

Teacher question: If the robot travels short, should counts per inch increase or decrease?

Technical notes

LinearOpMode. Code executes in a readable sequence, but no movement occurs before waitForStart returns.

Encoder reset. STOP_AND_RESET_ENCODER establishes zero, followed by RUN_USING_ENCODER before targets are created.

RUN_TO_POSITION. Target position supplies direction; use a positive magnitude for power. The controller uses encoder feedback, not floor localization.

Busy logic. OR between left and right isBusy waits until both finish. The timeout prevents indefinite driving if one side stalls.

Stop paths. Normal completion, Driver Station STOP, timeout, and the finally block all lead to zero power.

Calibration. Change constants only after repeated consistent measurements. Large left-right disagreement calls for mechanical inspection, not a single global scale factor.

Assessment evidence

- Correct target calculation
- Accurate code-flow explanation
- Unprompted completion of call-and-clear
- Three or more controlled trial records
- Reasonable calibration calculation, confirmation result, and exit explanation